

"Kestrel 1.5T: Catalyst efficiency below threshold from degraded converter"

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Area: bulletins

1 Condition: Catalyst Efficiency Below Threshold on Kestrel 1.5T

Kestrel 1.5T vehicles may illuminate the MIL with DTC P0420, indicating catalyst system efficiency below threshold on Bank 1. There are typically no driveability complaints; the fault is detected by the PCM comparing upstream and downstream oxygen sensor switching activity. The code most often sets after extended highway operation when the catalyst monitor completes. For background, see the Kestrel Emissions Service Manual :: Catalytic Conversion .

2 Cause: Degraded Catalytic Converter

The root cause is thermal aging and reduced oxygen-storage capacity of the catalytic converter, part number CAT-5500, which can no longer sustain the conversion efficiency the P0420 monitor requires. Oil or coolant contamination from an unrelated fault can accelerate this degradation, so confirm no upstream condition is present before condemning the converter. Identify the correct converter and its bank application using the Kestrel Emissions Service Manual :: Catalytic Converter (Bank 1 and Bank 2) . Monitor logic and pass/fail criteria are described under Kestrel Emissions Service Manual :: Catalyst Efficiency Faults .

3 Diagnosis: Downstream O2 Sensor Verification

Before replacing the converter, verify the downstream oxygen sensor O2D-5120 is responding correctly, since a lazy or biased sensor can mimic a P0420 catalyst fault. Run PROC-O2-SENSOR per the Kestrel Emissions Service Manual :: Upstream and Downstream Oxygen Sensors to confirm switching behavior, and replace O2D-5120 if out of specification using the Kestrel Emissions Service Manual :: Oxygen Sensor Replacement Procedure . Broader sensing context is provided in the Kestrel Emissions Service Manual :: Oxygen Sensing section.

4 Repair Procedure: Catalytic Converter Replacement

When the downstream sensor checks good and the catalyst monitor still fails, replace converter CAT-5500 following PROC-CAT. Perform the replacement per the Kestrel Emissions Service Manual :: Catalytic Converter Replacement Procedure , inspecting gaskets and flange hardware for exhaust leaks. After installation, clear P0420 and drive the vehicle through a full catalyst monitor cycle to confirm the readiness flag sets and efficiency returns above threshold.